



Zachary Treisman

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Technical Focus statistical modeling and predictive analytics; behavioral and user analytics; applied machine learning; experimental design and A/B testing; interpretable models.

Core Skills

- **Modeling & ML:** Predictive modeling, classification, regression, random forests, clustering, causal inference, hierarchical and mixed models
 - **Statistics:** Experimental design, A/B testing, hypothesis testing, behavioral segmentation, generalized linear models
 - **Programming:** R (tidyverse, ggplot2, glmmTMB, brms, randomForest); Python (pandas, NumPy, scikit-learn); SQL (BigQuery); Git; Unix/Linux
 - **Data:** Clinical EHR (MIMIC-IV), environmental/spatiotemporal, survey and administrative datasets
 - **Communication:** Translating complex analyses into actionable recommendations for non-technical stakeholders across domains
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Consulting & Applied Research

- **Data Scientist**, *ICU Extubation Outcomes: Caregiver Volume, Failed Extubation Rate, and Patient Survival* (joint with Andrew Hersh, MD, Pulmonary Critical Care). Built a BigQuery-to-R pipeline across MIMIC-IV v3.1. Caregiver failed extubation rate independently predicts 12-month survival. Nonlinear, monotonically decreasing relationship with survival floor, raising questions around the standard clinical benchmark. Three-level hierarchical modeling, including Bayesian methods and propensity score matching. Manuscript in development.
<https://github.com/ztreisman/ICU-Extubation-Outcomes>
- **Statistical Consultant**, *Ramah Navajo Chapter v. Salazar* (US Supreme Court, \$1B settlement). Designed sampling and estimation framework for unpaid tribal contract costs across tens of thousands of contracts spanning 19 years. Communicated findings to legal teams, accountants, and the Justice Department.
<http://www.scotusblog.com/case-files/cases/salazar-v-ramah-navajo-chapter/>
- **Statistical Consultant**, *WCU Salary Assessment Task Force*. Conducted pay equity analysis comparing university salaries to peer institutions, adjusting for cost of living and appropriate covariates. Built an R-Shiny interactive tool allowing leadership to explore the effects of different policy scenarios on salary equity outcomes in real time. Results initiated ongoing salary corrections across the institution.
- **Data Analyst**, *Gunnison Valley Winter Backcountry Recreation Study* (with US Forest Service). Modeled visitation patterns across user types (skiers, snowmobilers, mechanized, hybrid) against weather conditions, avalanche danger ratings, location, and day of week. Built negative binomial GLMs with AIC-selected interaction terms; revealed strong behavioral segmentation including week-end effects and season-to-season shifts. Delivered actionable insights and visualizations to public lands managers. Applied ML for automated image classification of trail camera data.
- **Data Analyst**, *San Juan National Forest Beetle and Fire Recovery Study* (with Jonathan Coop and Kyle Rodman). Built ML and statistical models from vegetation survey and satellite data to characterize forest recovery drivers after catastrophic fire and beetle infestation. Designed and analyzed treatment/control comparisons across spatially structured ecological data.
- **Independent Research**, *$\{3,7\}$ Hyperbolic Tiling Embedding Project*. Investigating the maximum rigid embedding depth of the $\{3,7\}$ hyperbolic tiling in $\mathbb{R}^3$. This surface grows exponentially but must fit in Euclidean space. Developed a physics and GNN-based ring-by-ring embedding pipeline (with a student team and Claude); exploring algebraic snap detection via quotients of the $\Delta(2,3,7)$ triangle group. Have a sketch of a proof that the maximum embedding depth is 5. This project has become a sustained experiment in AI-assisted mathematical research: context files are knowledge transfer artifacts raising concrete questions about what structures survive retrieval.
<https://github.com/ztreisman/237>

Academic Positions

- **Faculty & Honors Program Director, Western Colorado University.**
 - Primary academic statistician on campus; teach Statistical Modeling, Probability & Statistics, and Advanced Ecological Data Analysis; supervise student research projects in experimental design and applied modeling.
 - Direct an interdisciplinary honors program of ~150 students, managing a cross-disciplinary faculty team across departments university-wide; led cross-functional teams in the redesign of Western's General Education program, coordinating faculty, administration and staff to implement a university-wide structural change.
- **Postdoctoral Fellow, University of British Columbia.**
- **Visiting Lecturer, University of Washington.**
- **Visiting Fellow, Tata Institute of Fundamental Research.**

Prior Experience

- **Knowledge Engineer, Kindred Communications** (Microsoft Vendor). Built Bayesian network logic for the Windows help system using a custom probabilistic reasoning tool. Consulted with David Heckerman at Microsoft Research on probabilistic modeling questions.

Education

- **Ph.D. in Mathematics, University of Washington.**
- **B.A. in Mathematics, Reed College.**
- **Certificate in Data Mining, University of Colorado Boulder / Coursera.**
- **Santa Fe Institute Complex Systems Summer School.**
- **Budapest Semesters in Mathematics.**

Selected Publications & Projects

- **Ramah Navajo Chapter v. Salazar.** Statistical consultant, US Supreme Court case. \$1B settlement. <http://www.scotusblog.com/case-files/cases/salazar-v-ramah-navajo-chapter/>
- **The Boundary of the Visible.** *TSAIBERNETICS: transgenerational cybernetic art* — Wen-Ying Tsai, London Tsai, Ming Tsai. Laznia Centre for Contemporary Art.
- **A Young Person's Guide to the Hopf Fibration.** <https://arxiv.org/abs/0908.1205>
- **To the World's End/ A Circle Bundle Over a Circle.** *Bridges Stockholm 2018: Proceedings*, <http://www.londontsai.com/qr/worldsend.htm>
- **Rationally generated arcs.** *Journal of Geometry*, Volume 90, Numbers 1–2, 2008.